AMD Developer Hackathon ACT III

# Two Hands

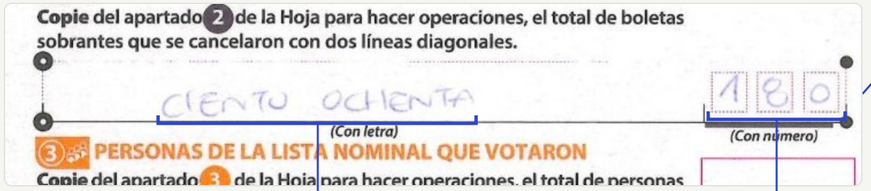
## Verifiable quality control for handwritten numbers.

OCR reads the number. Two Hands checks whether the whole form agrees.

For the institutes that run Mexico's vote counts and their auditors, and any handwritten form whose numbers must add up.

**UNUSED BALLOTS: TWO RECORDS OF ONE NUMBER**

| | |
|---|---|
| Official typed capture (PREP) | 480 |
| Gemma read | 180 |




In words: ciento ochenta (180)
the paper

In digits: 180
the paper




Tally sheet 080310B0100 (INE 2024), privacy-masked.

**1,270**
real handwritten tally sheets read by Gemma on AMD

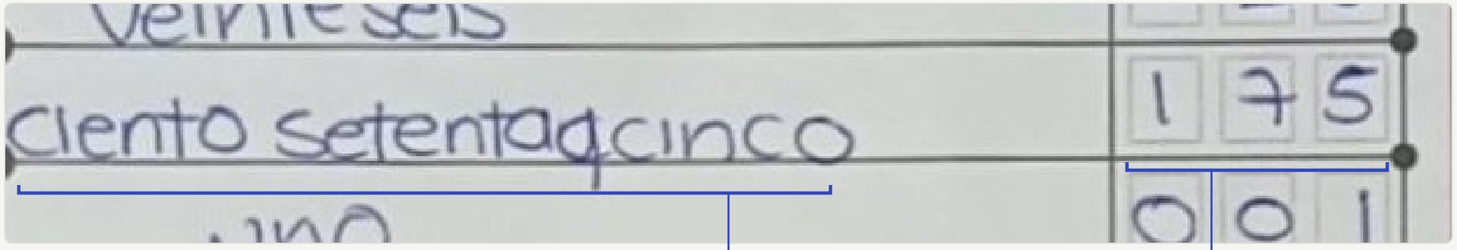
**91.6%**
field match, vs 62.84% for the reading the state bought

**27** of 28
capture errors the certified count reveals, caught (one state, pre-registered)

**0** of 484
accepted fields wrong on 100 fresh forms, gate fixed in advance

THE PROBLEM, FROM INE'S OWN FILES

# Two official records of the same paper disagree. Neither says which one is right.




In words: ciento setenta y cinco (175)
the paper

In digits: 175
the paper

Form 050815B0100, Option 07 (masked, cropped right of the logo column). One of the 52 on my random 300.

THE TWO TYPED RECORDS OF THIS FIELD

| | |
|---|---|
| Official typed capture (PREP) | 75 |
| Certified count | 175 |

The 75 is a typing error in the database, not on the paper.

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**6.5%**

of the 52,167 polling stations not recounted: the preliminary capture and the certified count disagree on at least one vote field

**19.2%**

of nationally captured tally forms fail their own row sum (26,336 of 137,196)

Double keying copies the paper faithfully but **cannot say which paper is wrong**. The photo can.

HOW IT WORKS

# The model reads. The paper's own arithmetic decides.

01
## Photo
Page located and privacy-masked before it is stored or sent

02
## Gemma reads
Gemma-4-31B-IT on AMD Token Factory reads every field twice: digits and words

03
## Arithmetic
words = digits · rows sum to TOTAL · section 5 = section 7 = TOTAL

04
## Accept or route
All hold: **accepted**. One fails: **a person** gets the field, with its photo

05
## Receipt
Each field a Merkle leaf; the root anchored with OpenTimestamps

**211298B0100**

INE 2024 tally sheet, type 2 · phone photo · Puebla · certified by comparison

sample: capture differs from certified

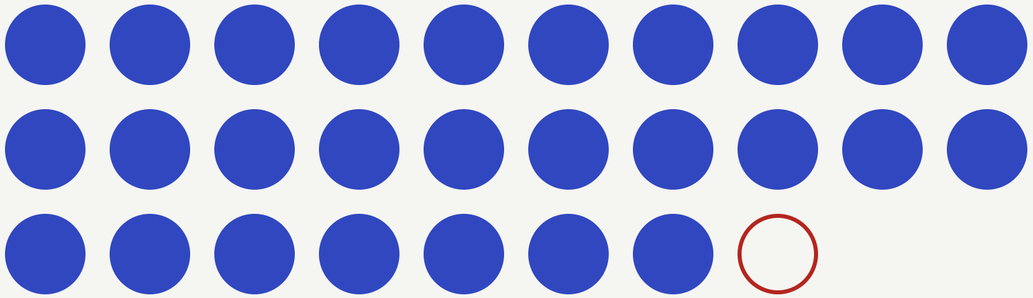
**Accepted: 0 fields to key**
Every identity on the page holds, so nobody keys this form.

| | | |
|---|---|---|
| ✓ | rows sum to TOTAL | 417 = 417 |
| ✓ | votaron + representantes = section 5 | 417 = 417 |
| ✓ | section 5 = section 7 | 417 = 417 |
| ✓ | section 7 = TOTAL | 417 = 417 |
| ✓ | words = digits on every field | 23 of 23 |

An accepted form on the site (211298B0100): every identity holds, so nobody keys it. A second reader, Qwen3.8-27B on AMD, never overrides Gemma: it can only withhold acceptance.

RESULTS · PRE-REGISTERED · INE'S PUBLIC DATA

# 1,270 real tally sheets read on AMD. 49 capture errors confirmed by the certified count.



Recall, Colima: each dot is a capture error the certified count reveals. Filled: caught. Outlined: missed. One state; not total recall.

**1,270**

handwritten tally sheets read by Gemma on AMD (289 of a random 300 and Colima's 982, one sheet in both)

**49**

capture errors confirmed by the certified count, each with its photo

**27** of 28

caught, 96% (Wilson 95% 82-99%), pre-registered

**0** of 484

accepted fields wrong on 100 fresh forms (gate v2, fixed before the draw; Wilson 95% upper bound 0.79%)

I re-read the AI-judged disputes myself, blind to its answers: **0 read differently**. Every crop is public.

AGAINST THE INCUMBENT · BAJA CALIFORNIA SUR, 2024 LOCAL ACTAS

# 91.6% field match, where the reading the state bought scored 62.84%.

Same per-field exact-match metric on Baja California Sur's 2024 local actas: mine on a pre-registered random 300, against the published values; theirs on all 2,020, against the manual capture.




Ours: 91.58% of 3,622 fields, Wilson 95% 90.63-92.44%. Theirs: 14,501 of 23,077 fields.

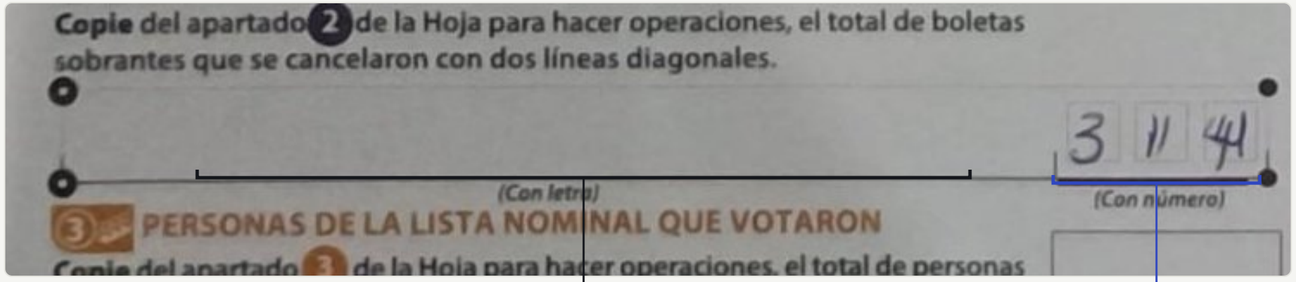
# Same 300 forms, same checks: Gemma accepts 1.8× the forms Qwen does, 4.6× cheaper, 2.8× faster.

| READER, ON AMD | FORMS ACCEPTED | WRONG OF ACCEPTED FIELDS | $ PER 1,000 FORMS | TIME PER READ |
|---|---|---|---|---|
| **Gemma-4-31B-IT** | **55 of 300** | **2 of 1,210** | **$0.14** | **87 s** |
| Qwen3.8-27B | 31 of 300 | 1 of 682 | $0.64 | 241 s |
| Both must agree, the paper's sums as referee | 89 of 300 | **0 of 1,958** | $0.78 | – |

Rows 1-2: the first gate (v1) on the pre-registered 300. Row 3: two readers, a lighter gate, a secondary analysis. Gate v2 on 100 fresh forms: slide 7. Source: app/data/bakeoff_300.json.
Time per read: wall clock in the 300 run (2 in flight, my own pacing). Measured separately on the shared endpoint, 4 in flight (12 requests): 4.8 forms/min, 32 s p50 and 67 s p95 per read; 9.6 output tokens/s per request (300 run). For an audit after the count: Colima's 982 sheets in about 3.4 hours (results/throughput_sweep.json).

**$0.14** **Colima, a whole state:** all 982 located presidential sheets, read by Gemma on AMD.

# Read 3114, paper 314. Gate v2, fixed in advance, sends it to a person.




Words line: blank
no second reading to compare

The paper: 314
in digits

Form 070212B0100, unused ballots (masked crop).

### THE BLIND SPOT

The model read **3114**. Unused ballots sit in no identity: no sum can catch a digit inserted here. The first gate would have accepted it.

### HELD OUT

Gate v2 (digits and words both read and equal) was fixed before 100 fresh forms were drawn. On them: **0 wrong of 484** accepted fields (Wilson 95% upper bound 0.79%).

### SCOPE

By design, an audit of the typed values: most forms go to a person, and the auditor sees only the disputed fields. Not an official INE product.

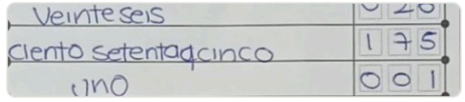
BUSINESS AT A GLANCE

# An independent check of the keyed values. $1 per sheet audited (proposed).

## Audit report: 6,358 fields checked, 267 disputed, each with its photo

PREP 2024, 300 pre-registered forms · 300 forms in the keyed file, 289 compared · audit `ab2b0555076b1` · 2026-10-17 · Merkle root `e56a8ef4f011b9e5` · fingerprint `c4788f2501814fd1`

3




In English: words: "ciento setenta y cinco" = 175 · boxes: 1 7 5

**050815B0100 · Option 07** STRONG

receipt

Judge this verdict yourself

| Keyed | Model digits | Model words |
|---|---|---|
| 75 | 175 | 175 |

- model digits and words agree (175)
- the keyed value breaks rows sum to TOTAL (258 vs 358); the model's value satisfies it

The auditor looks at **267 of 6,358 fields (4%)**, each with its photo.

The product: the audit report, strongest first, sealed by one Merkle root.

**Evolus:** each strong finding opens a case. Agent Intake clerk on Gemma-4-31B-IT, workflow Audit finding; a script, not the model, scores the evidence. 150 of 150 runs completed.

### BUYER

INE and the 32 state institutes, each with an outside auditor.

### WHY NOW

**6 June 2027**: the federal midterm and 32 concurrent local elections. First step: a **paid shadow-mode pilot** on one state's count.

### ONE STATE, PRICED

Morelos paid MXN 26.6M to run its 2024 PREP. My check of every sheet: about **MXN 143k**, 0.5% of that. The federal count, 169,005 sheets: about $169,000.

### NEXT

The check is the form's own arithmetic, so it isn't tied to elections. Proven on two form types: Mexico's tally sheets and UK rainfall ledgers (1850-1960), once typed by ~16,000 volunteers, no retraining. Next: tally sheets in at least 7 more countries that publish their images, and the monthly health-facility reports behind DHIS2, used in 70+ countries.

CHECK IT YOURSELF

# Every number re-derives from public data.

REPRODUCE

```
make eval
```

Recomputes every number offline and checks it against what the site shows.

VERIFY A RECEIPT

```
python -m twohands.verify_receipt
  receipt.json --crop crop.jpg
```

A changed value or crop fails.

CHECK THE TIMESTAMP

```
ots verify
results/sample_audit_root.txt.ots
```

The report's Merkle root, anchored in Bitcoin with OpenTimestamps.

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**Two Hands.** Verifiable quality control for handwritten numbers. `github.com/py-tr/twohands`